

Scale-Coupled Dynamics II: the pairing, the source law, and a torsion-free antisymmetric Ricci

Abstract

Scale-Coupled Dynamics (SCD) is a formalised framework in which the only primitive is a local scale and the geometry is its deformation. This note reports a revision pass carried out entirely inside the Lean 4 formalisation. Its results are of three kinds.

Structural. The founding statement $scale \times energy = constant$ needs only a group homomorphism; the commutative ring the development was built on was silently supplying a second, independent thing — a bilinear pairing for the geometry. Separating them consolidates the axioms from seven to five, and shows that the pairing was *never chosen*: the gravitational coupling was computed with a Euclidean trace while the observables live in a Lorentzian sector, joined by a single theorem whose hypotheses are mutually exclusive.

Corrective. Redoing the curvature identity with a signature shows the coefficient $\kappa = -2(n-1)\Delta$ survives unchanged, for a structural reason, but the operator it multiplies is the d'Alembertian rather than the Laplacian: the source law is hyperbolic, and Poisson's equation is its static limit. Two further claims lose their stated grounds while keeping their conclusions, and one claim is withdrawn.

Predictive. Off the commuting locus the Ricci tensor is *not symmetric*, with $Ric_{be} - Ric_{eb} = (n-2)[\sigma_b, \sigma_e]$, while the connection remains torsion-free and the scalar curvature does not move at all. Nothing classical — Newton's law, the coupling, cosmology — takes a quantum correction at any order; in particular there is no $\hbar G/r^3$ correction to the Newtonian potential. Finally, the usual four-constants–three-dimensions bookkeeping collapses to a single grading in which $\hbar$ and G carry the *same* weight, so that “are they the same magnitude?” becomes a well-formed question whose answer is one pure number — and the framework's long-quoted “one free pure number” is conditional on answering it.

All statements below are theorems in a Lean 4 development of 1275 theorems, every one of which passes `#print axioms` with no axiom beyond `propext`, `Classical.choice` and `Quot.sound`, and with no `sorry`.

Contents

1	Scope, and how to read this	2
2	The founding statement needs a group, not a ring	2
2.1	So what was the ring doing?	3
3	Axiom consolidation: seven to five	3
4	The source of the geometry cannot be a topological charge	4
5	The pairing was never chosen: two metrics	4

6	The repair: the coefficient survives, the operator does not	5
6.1	But the operator changes	5
7	Null characteristics, and what a scale wave is	5
7.1	The effect is quantised by the threshold spectrum	6
7.2	The breathing mode: expressible, but not radiable	6
8	Off the commuting locus: an antisymmetric Ricci without torsion	6
8.1	What this predicts	7
9	Constants and dimensions	8
9.1	Where e goes	8
10	What remains open	9
11	Method	9

1 Scope, and how to read this

SCD posits a dimensionless local log-scale σ over a ring with n commuting derivations, and derives geometry as the deformation of that scale rather than as an independent metric structure. This note does not re-derive the framework. It reports what a deliberate examination of its *mathematical form* produced, and it is written so that each claim can be checked against a named Lean declaration.

Three conventions are used throughout, and they are not decoration.

- **Claimed** means proved in the formalisation, with the declaration named.
- **Identified** means an interpretive step that the formalisation records as an input rather than a consequence.
- **Withdrawn** means a claim previously made in this development and now retracted, with the reason given.

Section 10 collects what remains open, including one item this pass deliberately declined to pursue.

2 The founding statement needs a group, not a ring

The framework's founding statement is $scale \times energy = constant$, encoded by two axioms: A2 supplies a log-scale σ with $s = e^\sigma$ a unit, and A3 sets $\varepsilon = s^{-1}$.

The theorem saying what this means had never been written.

Theorem 2.1 (Reduction.logDeriv_mul). *For units s, t of the ring and each direction i ,*

$$\logDeriv(st)_i = \logDeriv(s)_i + \logDeriv(t)_i.$$

The log-derivative $s^{-1}\partial_i s$ is a *group homomorphism* from the units into the additive group of gradients. Two consequences follow at once.

Corollary 2.2 (Reduction.logDeriv_inv). $\logDeriv(s^{-1}) = -\logDeriv(s)$.

This *is* A3. That the energy's gradient is minus the scale's is inversion in a group, not an independent postulate.

Definition 2.3 (`Reduction.ScaleHom`). A group G , an abelian group M , and a map $D: G \rightarrow M^n$ with $D(st) = Ds + Dt$.

Every scale algebra furnishes one (`Reduction.ofScaleAlgebra`). The signature contains *no ring, no commutativity, and no multiplication*. That is the whole of what the founding statement requires.

2.1 So what was the ring doing?

Theorem 2.4 (`Reduction.sig_add`, `Reduction.gradsq_add`). *The gradient is additive in the log-scale, while*

$$\text{gradsq}(\sigma + \tau) = \text{gradsq } \sigma + \text{gradsq } \tau + 2 \sum_i \text{sig } \sigma_i \cdot \text{sig } \tau_i.$$

The axiom is linear in the gradient; the geometry is quadratic; and the failure of additivity is exactly a *symmetric bilinear pairing*. Every geometric object in the development is built from that pairing.

A commutative ring supplies the homomorphism and the pairing in one move. That is why it was chosen, and it was never recorded that these are two commitments rather than one.

The sharp form of the concern: in a ring, the multiplication defining the unit group and the multiplication supplying the pairing are the *same operation*, and nothing in $\text{scale} \times \text{energy} = \text{constant}$ requires that.

3 Axiom consolidation: seven to five

Three results turn a long-standing goal into a theorem.

Proposition 3.1 (`Consolidation.A3_is_inversion`). *The duality relation is `Units.inv_mul` and $\varepsilon := s^{-1}$ is a definition. A3 names a definition and asserts a group law.*

Theorem 3.2 (`Consolidation.d_s_iff_exact`). *For a unit s and an element σ ,*

$$(\forall i, \partial_i s = s \cdot \text{sig } \sigma_i) \iff (\forall i, \text{logDeriv}(s)_i = \text{sig } \sigma_i).$$

A2's derivation condition holds exactly when the unit's log-derivative is *exact*. So A2 is a unit plus a potential; A4' (one unit per direction) already supplies the unit; and no geometric result needs the potential (`Amendment.same_unit_same_geometry`).

The consolidated list is

$$A1 \cdot A2^* \text{ (one family of units)} \cdot A5 \cdot A6 \cdot A6'' \cdot A7,$$

recorded as `Consolidation.consolidation`. The reduction is not claimed to be *forced*: a framework may keep a potential primitive. What has changed is that the potential is now known to be the only surplus.

4 The source of the geometry cannot be a topological charge

A6'' reads $\Delta\sigma = \Delta \cdot \nu$ and was presented as an *index* law, with ν a topological charge. The development tabulated three candidate integers. All three are eliminated by one theorem.

Theorem 4.1 (`DegreeSource.additive_nonneg_on_group_is_trivial`). *Let G be a group and let $f: G \rightarrow \mathbb{R}$ satisfy $f(ab) = f(a) + f(b)$ and $f \geq 0$. Then $f \equiv 0$.*

The proof is three lines: $f(1) = 0$, then $f(a) + f(a^{-1}) = 0$ with both terms non-negative.

Additivity is what makes a source an index (`Flux.flux_combine`); non-negativity is what universal attraction requires (`Attraction.attraction_from_counting`). Hence:

No group-valued charge can be a universally attractive additive source. Every topological charge here is group-valued — π_1 and π_2 are groups — so the source of the geometry cannot be topological at all.

The argument is sharp rather than over-strong. `DegreeSource.monoid_source_exists` exhibits an additive, non-negative, non-zero map on $\mathbb{N}$: it is *inverses* that do the killing. What survives is a threshold count, a cardinality with no anti-content. Two external checks: real gravity's source is not topological and mass is non-negative; and electric charge *is* signed and *does* source an interaction, which the theorem permits because that interaction is not universally attractive.

Remark 4.2 (Consequence for A6''). The coefficient in $\kappa = -2(n-1)\Delta$ is algebraic in ν and is untouched. What is touched is A6''s standing: its authority came from ν being a topological index. A6'' keeps its consequences and loses its index reading; it is a postulate of the same standing as the rest of the axioms.

5 The pairing was never chosen: two metrics

The framework computes curvature in two places.

- `Conformal.RscBare` = $\sum_b \text{Ric}_{bb}$, described in its own documentation as the δ -trace of Ricci, on the metric $e^{2\sigma}\delta$. This is where $\kappa = -2(n-1)\Delta$ is derived, so κ is a **Euclidean result**.
- `Diagonal.Ric` on the metric $\text{met}\eta w$, with the signature carried as a parameter satisfying $\eta^2 = 1$. The development's own chain table runs the *entire* gravity chain through this sector, and the key cancellation requires $\eta_t\eta_r = -1$.

The single theorem joining the two, `Anisotropic.ChrDir_of_isotropic`, scopes itself in its own documentation to *Euclidean signature*, with hypothesis $\eta \equiv 1$.

Proposition 5.1 (`TwoMetrics.euclidean_excludes_lorentzian`). *If $2 \neq 0$ in the ring, then $\eta \equiv 1$ and $\eta_t\eta_r = -1$ cannot both hold.*

The only bridge between the sectors holds exactly where the physics does not. κ and the observables have never been shown to be about the same geometry.

`TwoMetrics.delta_trace_ne_eta_trace` shows the difference does not cancel: on a Lorentzian pair the two traces read 2 and 0.

Remark 5.2 (A correction to the record). The framework’s summary of what rests on the axioms alone listed “the Lorentzian signature”. What is derived is a *codimension-one splitting* of the directions from the drift. A splitting is a flag; a signature is a property of a form. That file’s own closing paragraph already granted that the timelike sign is not derived.

6 The repair: the coefficient survives, the operator does not

The curvature identity was redone from the connection upward with a signature, since inserting η into the final trace alone would not be the same geometry. The connection used is the one the development already had for the anisotropic sector, at the isotropic locus with η left general: a case that existed and had never been computed.

Theorem 6.1 (`EtaTrace.RscE_eq`).

$$R^\eta = -2(n-1)\square\sigma - (n-1)(n-2)|\nabla\sigma|_\eta^2,$$

where $\square\sigma = \sum_i \eta_i \sigma_{ii}$ and $|\nabla\sigma|_\eta^2 = \sum_i \eta_i \sigma_i \sigma_i$.

Identical coefficients. `EtaTrace.A6’_from_index_eta` then returns $\kappa = -2(n-1)\Delta$.

The reason is structural rather than fortunate. In the two-connection contraction two η -carrying terms cancel identically, the second carrying $\eta_e^2 = 1$; every surviving η is either squared or pinned by a δ that forces its indices equal. The signature cannot reach the coefficient.

6.1 But the operator changes

Theorem 6.2 (`Hyperbolic.lapE_lorentzian_split`). *With $\eta_t = -1$ and $\eta_i = 1$ for $i \neq t$, $\square\sigma = \sum_{i \neq t} \sigma_{ii} - \sigma_{tt}$.*

κ is safe; the operator it multiplies is the d’Alembertian. The source law $\square\sigma = \Delta \cdot \nu$ is **hyperbolic**, not elliptic — a sourced wave equation. Newton is recovered as its static limit, where $\sigma_{tt} = 0$ and the d’Alembertian is the Laplacian, and the framework’s gravity results are all computed under staticity.

`Hyperbolic.lap_sub_lapE` measures the gap as $\sum_i (1 - \eta_i) \sigma_{ii}$, zero only for a Euclidean signature. Consequently the framework’s Poisson-limit theorem should be read as *scoped to static configurations*. Nothing computed from it changes; what changes is what it is a limit of.

7 Null characteristics, and what a scale wave is

A hyperbolic law has characteristics. Since no integration theory is available, the wave is treated algebraically: a *wave profile* is a scale whose Hessian is rank one along a covector k , that is $\sigma_{ij} = k_i k_j m$.

Theorem 7.1 (`Hyperbolic.lapE_of_waveProfile`, `Hyperbolic.vacuum_wave_is_null`). $\square\sigma = \text{bareForm}(\eta, k) \cdot m$, and in vacuum with m a unit, $\text{bareForm}(\eta, k) = 0$.

That bare form is the framework’s *own* quadratic form, defined kinematically as the locus where the measured form vanishes. Nothing in the development had said that anything *travels* along it.

Scale disturbances propagate on the framework's own null cone: the dynamical characteristic cone and the kinematic null cone coincide.

7.1 The effect is quantised by the threshold spectrum

Content in SCD is what a probe at a given scale resolves, namely $\text{resolved}(\mu, t) = \{k \mid \mu_k \leq t\}$.

Theorem 7.2 (`Directional.resolved_eq_iff_no_threshold_between`). *For $t \leq t'$, $\text{resolved}(\mu, t) = \text{resolved}(\mu, t')$ if and only if no μ_k lies in $(t, t']$.*

A scale wave has no continuous observable. It acts only where it crosses a threshold, so half the amplitude gives *none* of the effect unless a threshold lies in the interval. A scale wave is a wave of emergence.

7.2 The breathing mode: expressible, but not radiable

The development contained a theorem excluding a breathing (scalar) polarization, justified by the claim that a uniform rescaling is a fiducial shift and so unobservable by A5.

Proposition 7.3 (`Breathing.nonconstant_scale_is_expressible`, `Breathing.uniform_across_directions_still_fiducial_invariance`). *Fiducial invariance quantifies over one constant added at every point, and the difference pattern compares points. A breathing wave is uniform across directions and varies across points, and is separated by a fiducial-invariant functional.*

So A5 does not remove the breathing mode, and the null-cone result exhibits it. The conclusion nevertheless survives, on a ground the framework already had: a scalar disturbance is sourced by the monopole; a conserved source's monopole is frozen (`Waves.monopole_does_not_radiate`); and a non-conserved source admits *no field equation at all* (`Dynamics.no_field_equation_of_nonconserved`).

The breathing mode is expressible but cannot be radiated by any source for which a field equation can be written. Same prediction, different ground, and a narrower scope: a free or primordial breathing wave is excluded by nothing here.

Remark 7.4 (Withdrawn). An earlier statement of this pass — *no interferometer can see a scale wave* — is withdrawn. The cited theorem takes a *single* scale pattern and compares two null directions at one point; it excludes finding a *static* anisotropy. A wave differs between the arms, and between emission and return. What survives is that a static scale pattern is invisible to an interferometer at any precision.

8 Off the commuting locus: an antisymmetric Ricci without torsion

Every quantitative result in the development is computed over a commutative ring, where the adjoint action vanishes identically and the quantum correction is zero by construction. This section leaves that locus.

The non-commutative Riemann tensor is known: curvature is still the Kulkarni–Nomizu product of the bare bookkeeping with the scale deformation, plus one correction built from commutators of scale gradients. Contracting it:

Proposition 8.1 (QuantumRicci.sum_RmCorr_contract). *The Riemann correction vanishes in the Ricci contraction: its two Kronecker terms cancel term by term at $c = a$.*

Theorem 8.2 (QuantumRicci.two_Ric_eq_nc). $2 \text{Ric}_{be} = (2 - n) D_{be} - \delta_{be} \text{tr } D$, with no extra term.

Yet the deformation tensor is itself not symmetric off the locus, and therefore:

Theorem 8.3 (QuantumRicci.two_Ric_antisymm).

$$\boxed{\text{Ric}_{be} - \text{Ric}_{eb} = (n - 2) [\sigma_b, \sigma_e]}$$

Theorem 8.4 (QuantumRicci.two_RscBare_eq_nc). $2R = (2 - 2n) \text{tr } D$, and $\text{tr } D$ is built from σ_{aa} , $\sigma_a \sigma_a$ and $|\nabla \sigma|^2$ — every term a square.

The entire quantum correction to the geometry is the antisymmetric part of Ricci. The scalar curvature does not move at all.

8.1 What this predicts

An antisymmetric Ricci with a torsion-free connection. The connection is symmetric in its lower pair whether or not the values commute (Conformal.Chr_symm, proved over a general ring), so there is *no torsion*. In ordinary differential geometry $\text{Ric}_{[ab]} = 0$ for a torsion-free connection, by the first Bianchi identity — and that proof uses commutativity of the *values*, not only torsion-freeness. This is therefore not Einstein–Cartan, which buys the same antisymmetry by giving the connection torsion.

Nothing classical moves, at any order. The source law, κ , the Newtonian limit and every cosmological statement are exactly their commutative selves. In particular there is **no $\hbar G/r^3$ correction to the Newtonian potential**, of the kind effective-field-theory treatments of gravity produce, and **no running of G** . This is a flat disagreement with the standard expectation, and it is exact rather than small.

It vanishes in two dimensions, by the same factor $(n - 2)$ that governs conformal geometry throughout.

Where it can appear. An antisymmetric Ricci in a field equation requires an antisymmetric source, and the physical antisymmetric source is spin. Independently, NCConformal.quantum_correction_is_a_w holds by **rf1**: the correction *is* the framework’s rotational label applied to the scale gradient pattern. *This is an identification, not a theorem*: no field equation for the antisymmetric half is written, and no magnitude is claimed.

And it does not inherit the pairing problem. By Micro.quantum_correction_signature_independent the signature enters the deformation tensor only symmetrically, so the antisymmetric part — where the whole correction lives — is the same for every η . Unlike κ , this prediction never depended on the pairing that was never chosen.

Usual	Here
L, T, M	one grading: weight under $\sigma \mapsto c\sigma$
c	not a constant — one cone; measured speed one is a theorem
$\hbar$	a weight-one scale step; an identification, not a construction
G	a weight-one coefficient Δ
e , i.e. α	weight zero — already pure, and structurally so
Planck units	one axis, so one magnitude fixes everything
pure numbers	$\Delta\rho$ and $\hbar\rho$: <i>two</i> , collapsing iff $\Delta = \hbar$

Table 1: The dimensional bookkeeping, translated.

9 Constants and dimensions

The usual bookkeeping has three base dimensions L, T, M and four constants $c, \hbar, G, e$. Here there are no dimensions at all: what replaces dimension is a single $\mathbb{Z}$ -grading, the weight under $\sigma \mapsto c\sigma$.

The collapse is structural rather than a choice of units. Time is a *direction*, told from space by a sign and not by a dimension. Light crosses at measured speed one in every direction, because the scale sets rods and light times alike, so $c = 1$ is a theorem. And A3, being a group law (§3), places mass on the same axis with the opposite sign.

The magnitudes on that axis are Δ (weight 1), the scale step identified with $\hbar$ (weight 1, since a step is a difference of log-scales), and the threshold density ρ (weight -1).

Theorem 9.1 (`Constants.two_pure_numbers`, `Constants.collapse_iff`). *Both $\Delta\rho$ and $\hbar\rho$ are weight zero, and they coincide if and only if $\Delta = \hbar$.*

There are *two* free pure numbers, not one, unless G and $\hbar$ are the same magnitude — and nothing in the development says they are.

The framework’s own files already record that “ G and $\hbar$ share an origin” is conditional on a step not taken, and that “ $\hbar$ is that step” is an identification rather than a construction. The two statements had never been read against each other.

9.1 Where e goes

The fine-structure constant is weight zero: a gauge coupling’s value does not depend on how σ is labelled. So it is a pure number *structurally*. And the asymmetry has a reason: the gauge law relates weight-zero quantities to weight-zero quantities, while gravity’s relates a weight-zero *count* to a weight-one *scale*.

Exactly one law crosses the grading. “Why is α dimensionless and G not” stops being a fact about dimensions and becomes a fact about which law crosses the grading.

Remark 9.2 (A question the usual picture cannot ask). $[G] = L^3M^{-1}T^{-2}$ and $[\hbar] = ML^2T^{-1}$ have different dimensions, so *are G and $\hbar$ the same magnitude?* is not a well-formed question there. Here they carry the same weight, the question is well-formed, and its answer is a single pure number.

10 What remains open

The directional source law. $A6'$ reads $e^{2\sigma}R = \kappa\rho$, and $e^{2\sigma}$ is *one* conformal factor. With a scale per direction there is no single factor: `Directional.dirTrace_factors_iff` shows the factor comes out only when every directional unit agrees. So $A6'$'s *form* presupposes an isotropic scale while $A4'$ supplies a directional one. A replacement is written (`Breathing.DirSource`) on the true scalar curvature — $A4'$ supplies exactly the invertibility the bare form was contorted to avoid — but the curvature identity that would let it produce a number does not exist off the isotropic locus. Until it does, κ and the observables meet only where the scale is isotropic.

The quantum correction has never been carried. Section 8 is the first computation off the commuting locus. Every number in the development is computed on the isotropic, commuting locus, and both unfinished sectors are off it.

$\Delta = \hbar$: **deliberately left open.** Nothing in the axioms relates the source coefficient to a scale shift, and the scale shift itself is not built from the axioms — it requires a ring endomorphism and a unit that nothing supplies. The identification would have to come from outside. It is recorded as an open question, and the free-number count reads *two*.

And one number that cannot be computed. Predictions are informationally disjoint from inputs, and counts cannot fix a unit; $\Delta\rho$ is a *permanent* input. The framework therefore predicts ratios and structures, not magnitudes, and the familiar arcsecond figures are weight-zero statements evaluated on measured inputs.

11 Method

Three times in this pass a result was kept while its stated ground was replaced: $A6''$ keeps its consequences and loses its index reading; the Poisson limit keeps its numbers and becomes a static limit; the breathing exclusion keeps its conclusion and changes its argument. Twice a file's own self-correction caught an error that no theorem would have. Once a claim made in this pass was withdrawn one section later.

The rate is the thing to read, not the instances. A formalisation makes the theorems reliable; it does not make the *prose around them* reliable, and most of what was corrected here lived in the prose — in what a true theorem was being quoted as showing.